

Analyzing Advertisements

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MIS370-1: Web Analytics

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Analyzing Advertisements

1) Provide the code and output for the header displaying the first five rows.

Figure 1

First Five Rows of the Advertising Dataset

```
[1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
sns.set()

[2]: ads = pd.read_csv("Advertising.csv", index_col = 'Date')
ads.head()
```

	Products	Web	Newspaper	Radio	TV
Date					
01/01/2018	Mobile	230100	69200	37800	22100
01/02/2018	Mobile	44500	45100	39300	10400
01/03/2018	Mobile	17200	69300	45900	9300
01/04/2018	Mobile	151500	58500	41300	18500
01/05/2018	Mobile	180800	58400	10800	12900

Note. This figure shows the first five observations after importing the dataset with pandas and setting Date as the index column. The preview confirms that the data loaded correctly. The steps for this exercise were adapted from Activity 2.01 in Baig et al. (2021).

2) Provide the code and output for the information about the data.

Figure 2

Information Summary

```
[3]: ads.info()

<class 'pandas.core.frame.DataFrame'>
Index: 200 entries, 01/01/2018 to 07/06/2019
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype  
---  -
0   Products    200 non-null    object  
1   Web         200 non-null    int64   
2   Newspaper   200 non-null    int64   
3   Radio       200 non-null    int64   
4   TV          200 non-null    int64   
dtypes: int64(4), object(1)
memory usage: 9.4+ KB
```

Note. The info() method shows that the dataset contains two hundred entries from 1/1/2018-7/6/2019. It includes five columns with no missing values. Four columns are integers while Products categorical. The steps for this exercise were adapted from Activity 2.01 in Baig et al. (2021).

3) Provide code and output for the statistics.

Figure 3

Descriptive Statistics

```
[4]: ads.describe()
```

```
[4]:
```

	Web	Newspaper	Radio	TV
count	200.000000	200.000000	200.000000	200.000000
mean	147042.500000	30554.000000	23264.000000	14022.500000
std	85854.236315	21778.620839	14846.809176	5217.456566
min	700.000000	300.000000	0.000000	1600.000000
25%	74375.000000	12750.000000	9975.000000	10375.000000
50%	149750.000000	25750.000000	22900.000000	12900.000000
75%	218825.000000	45100.000000	36525.000000	17400.000000
max	296400.000000	114000.000000	49600.000000	27000.000000

Note. The describe() method summarized numerical columns across two hundred entities. Web has the highest mean viewership (147,042.5) and maximum (296,400). Radio shows the lowest minimum (0). These values suggest that web campaigns generate more views compared to traditional media outlets. The steps for this exercise were adapted from Activity 2.01 in Baig et al. (2021).

4) Provide code and output for unique products.

Figure 4

Unique Product Categories

```
[5]: ads['Products'].unique()
```

```
[5]: array(['Mobile', 'Electronics', 'Laptops'], dtype=object)
```

Note. The unique() method applied to the Products column shows three distinct categories: Mobile, Electronics, and Laptops.

5) Provide code and counts for the products.

Figure 5

Count of Product Categories

```
[6]: ads['Products'].value_counts()
```

```
[6]: Mobile      129
     Electronics  53
     Laptops     18
     Name: Products, dtype: int64
```

Note. The value_count() method applied to the Products column shows that Mobile has the highest number of records (129), followed by Electronics (53), and Laptops (18). The steps for this exercise were adapted from Activity 2.01 in Baig et al. (2021).

6) Provide code and output for the total views across categories.

Figure 6

Total Views Across Product Category

```
[7]: ads.groupby('Products')[['Web', 'Newspaper', 'Radio', 'TV']].sum()
```

```
[7]:
```

	Web	Newspaper	Radio	TV
Products				
Electronics	7734400	1500100	1161200	698300
Laptops	2711700	379200	374000	257800
Mobile	18962400	4231500	3117600	1848400

Note. These grouped totals show that Web consistently generates the highest viewership across all product categories with Mobile being the largest overall Web exposure (18,962,400). Newspaper and Radio campaigns also contribute, though TB views are comparatively low. This suggests that Web is the dominant channel across all categories. The steps for this exercise were adapted from Activity 2.01 in Baig et al. (2021).

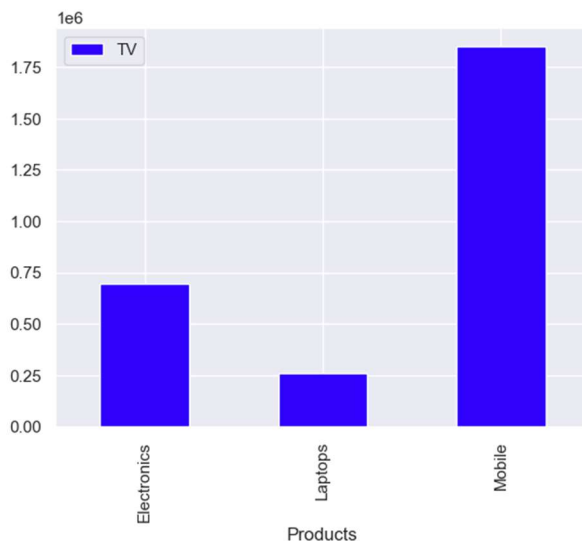
7) Plot a bar chart between Products and TV (this is your ‘y’). Use “blue” for the attribute color. Provide your code and output.

Figure 7

Bar Chart for TV Viewership

```
[8]: ads.groupby('Products').sum().plot(kind = 'bar', y = 'TV',color='blue')
```

```
[8]: <AxesSubplot:xlabel='Products'>
```

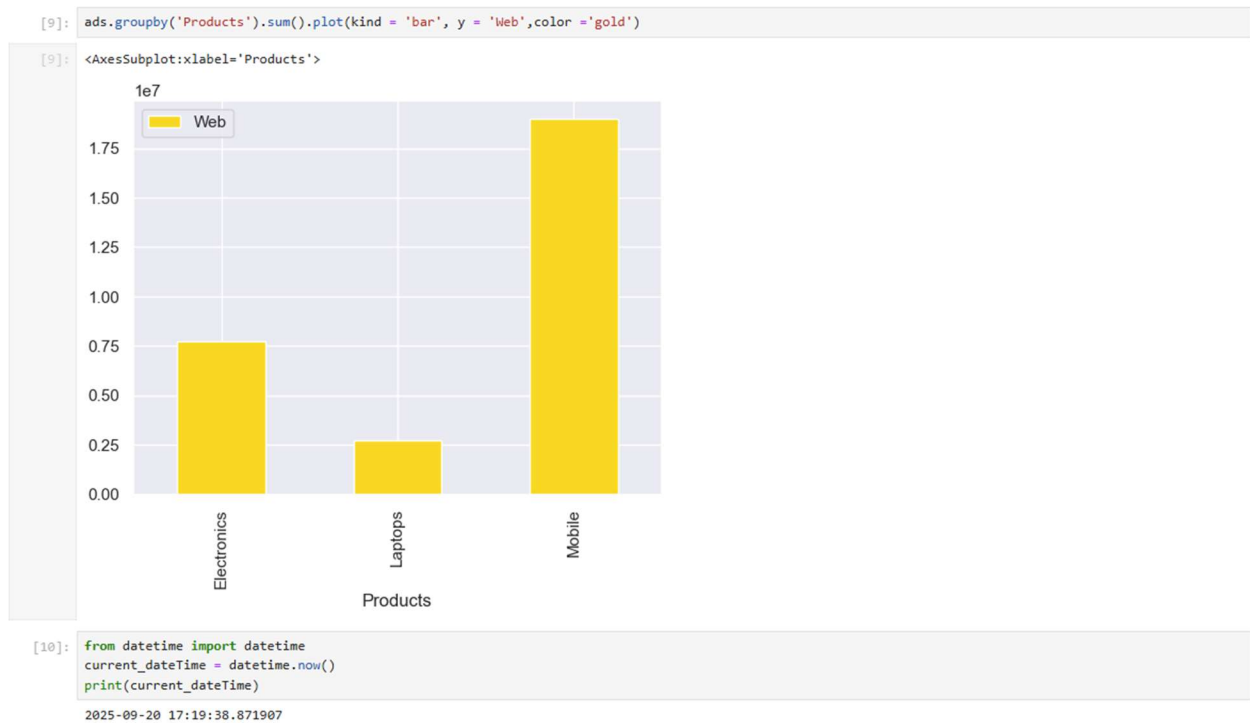


Note. This bar chart shows that Mobile products dominate TV advertising with the highest total viewership, followed by Electronics. Laptops receive the lowest TV exposure. The plot() function was applied to the grouped dataset with the attributes set to blue. The steps for this exercise were adapted from Activity 2.01 in Baig et al. (2021).

8) Plot a bar chart for which products have the lowest viewership on the web (this is your ‘y’). Use “gold” for the attribute color. Provide your code and output.

Figure 8

Bar Chart of Web Viewership



Note. The bar chart shows that Mobile has the highest Web viewership while Laptops remain the lowest. The plot() function was applied to the grouped dataset with the attributes set to gold. The steps for this exercise were adapted from Activity 2.01 in Baig et al. (2021).

9) Provide your observations of the questions listed in the activity.

Question	Observation
What are the unique values present in the Products column?	Mobile, Electronics, & Laptops
How many data points belong to each category in the Products column?	Mobile – 129 Electronics – 53 Laptops – 18
What are the total views across each category in the Products column?	Electronics ~ 7.73M Web, 1.5M Newspaper, 1.16M Radio, and 0.7M TV Laptop ~ 2.71M Web, 379k Newspaper, 374k Radio, and 258k TV Mobile ~ 18.96M Web, 4.23M Newspaper, 3.12M Radio, and 1.85M TV
Which product has the highest viewership on TV?	Mobile
Which product has the lowest viewership on the web?	Laptops

10) Provide a brief summary paragraph (100 words) of how you believe this will inform decisions.

The analysis of the Advertising dataset highlights how different media channels contribute to product viewership. The results show that Web campaigns dominate across all categories with Mobile products generating the highest overall exposure. Laptops consistently performed the worst across all categories with Radio being its worst product. These insights suggest that advertising budgets should emphasize Web campaigns while strategically allocating resources to TV and Newspapers for Mobile products. The company could potentially re-evaluate investments for Laptop products. By using statistics, counts, and visualizations, marketers can better align channel selection with product demand. These data-driven decisions can optimize advertising effectiveness, enhance ROI, and inform future campaign strategies.

Reference

Baig, M. R., Govindan, G., & Shrimali, V. R. (2021). *Data Science for Marketing Analytics* (2nd ed.). Packt Publishing. <https://reader.yuzu.com/books/9781800563889>